STPS: Spatio-Temporal Proactive Scheduling for Spiking Neural Networks on Compute-in-Memory Clusters

Your N. Here
Your Institution

Second Name
Second Institution

Abstract

Spiking Neural Networks (SNNs) deployed on Compute-in-Memory (CIM) clusters present a scheduling problem that does not appear in conventional DNN serving: each model's network-on-chip (NoC) traffic is both *input-dependent* and *time-varying*, so a tenant that looks idle on average can saturate a card's NoC during a microsecond-scale spike burst. A formulation that schedules every spike is a time-dependent mixed-integer non-linear program (MINLP) and is intractable at admission time; reactive runtime mitigation only observes congestion once the network has already saturated.

We present **STPS**, an admission-time scheduler that decouples what can be known offline from what must be decided online. An offline profiler reduces each SNN to four *fingerprints* extracted once from a Discrete-Time Dynamic Graph (DTDG) over a calibration trace; the two-stage online scheduler is driven by three of them — traffic timeline $E^{(t)}$, burstiness β, and mean active components $\overline{K}$ — with a per-population hotspot indicator recorded for future use. On every admission, the scheduler consumes the fingerprint to choose a card in closed form and, in $O(D_{\max} \cdot H)$, a start offset that fits the task's traffic timeline into a residual valley of the card's forecast.

In simulation against four standard cluster schedulers (RR, BestFit, DRF [3], P2C) on a mixed-fingerprint workload built from synthetic templates and SpikingJelly [2] traces of Spikformer [18], QKFormer, and SpikingResFormer, STPS is the only scheduler that consistently lowers NoC congestion — 5–8 % lower congestion ratio and 7–9 % lower mean queue wait — in every arrival/scale cell we test, at the cost of 1–2 % throughput. On a 16-card cluster, STPS holds the worst-to-best card load ratio at 12× while baselines develop a 14–24× tail. Stage-A and Stage-B ablations show the two stages address complementary axes: macro-dispatch flattens the across-card distribution; phase-shift collapses cross-baseline NoC congestion by more than 20× for under one tick of average added delay. The trade STPS makes shows up clearly at the long tail of per-task delay — STPS adds 6–12 ticks at $p99$ in exchange for the cluster-wide gains above — a regime we characterise explicitly in Section 7.5.

1 Introduction

Energy-efficient deep learning has rekindled interest in Spiking Neural Networks (SNNs) on neuromorphic Compute-in-Memory hardware. Production-class platforms now span academic and industrial designs — TrueNorth [10], Loihi and Loihi-2 [1, 13], SpiNNaker-2 [9], and Darwin3 [8] — and their event-driven execution model converts a model's idle neurons into idle silicon. The promise is attractive enough that a *Neuromorphic Computing-as-a-Service* cluster — many tenant SNNs sharing one bank of CIM cards — is a plausible deployment target for the next generation of neuromorphic systems.

Scheduling on such a cluster looks superficially like a DNN serving problem and is fundamentally different. A DNN model has a deterministic compute graph: at admission, a placement decision based on FLOPs, memory, and bandwidth is correct for the entire run, and predictable execution is the design goal of recent serving stacks such as Clockwork [5]. An SNN's per-tick activity, by contrast, is a function of the input. A vision classifier on a quiescent frame fires a handful of spikes; the same model on a high-contrast frame can inject orders of magnitude more traffic into the NoC. Even within one inference, spikes arrive in narrow temporal bursts driven by neural dynamics rather than uniformly across the inference window. Two consequences follow. First, capacity-based placement — best-fit on free cores, dominant-resource fairness on weighted resources [3], power-of-two-choices on instantaneous load [11] — leaves cards that are spatially balanced and temporally hot, because the bursts of co-located tenants align and the per-card NoC saturates. Second, treating each spike as a scheduling unit is infeasible: at admission the scheduler does not yet know which spikes will fire, and even if it did the resulting MINLP would not solve in the per-task latency budget of a serving system.

A pragmatic scheduler must therefore reason about *distributions* of spike activity, not individual spikes. The information

that distinguishes a "quiet" task from a bursty one is statistical, available before the task arrives, and small enough to consult in $O(1)$. We make this concrete: a Discrete-Time Dynamic Graph (DTDG) view of an SNN trace yields four compact fingerprints — the traffic timeline $E^{(t)}$, global burstiness β, mean active components $\overline{K}$, and a per-population hotspot indicator — of which the first three suffice to drive admission-time placement and timing decisions without re-examining the underlying weight tensor (the hotspot indicator is recorded for a future spatial-splitting extension).

The resulting scheduler, **STPS** (Spatio-Temporal Proactive Scheduling), takes these fingerprints as input. A macro-dispatch stage scores each card with a closed-form weighted sum of fragmentation and burstiness pressure, picking the card whose fragmentation matches the task's topology and whose forecast burst-density tolerates the task's β. A phase-shift stage then searches at most $D_{\max}$ start offsets and commits the offset whose addition to the card's forecast minimises the peak ($O(D_{\max} \cdot H)$). The two stages address orthogonal failure modes — spatial fragmentation and temporal collision — and are the minimum mechanism that closes the gap between capacity-based DNN scheduling and SNN execution dynamics.

Contributions.

- A *workload fingerprint* that distils an SNN's spatio-temporal injection profile into four scalars/vectors derivable from a calibration trace, sufficient to support $O(1)$ admission-time decisions (Section 5.2).
- A two-stage online scheduler that consumes the fingerprint to dispatch tasks across cards and to shift their start offsets, with both stages reducing to closed-form scoring or a bounded search over $D_{\max}$ offsets (Sections 5.3.1 and 5.3.2).
- An evaluation on real and synthetic SNN traces showing that STPS lowers NoC congestion by 5–8 % and queue wait by 7–9 % across arrival modes, holds the 16-card tail load at 12× versus baselines' $14-24\times$, isolates the contribution of each stage, and explicitly characterises the long-tail delay cost the scheduler pays (Section 7).

The rest of the paper is organised as follows. Section 2 explains why both capacity-based and per-spike scheduling fail on SNN/CIM clusters; Section 3 positions STPS against prior work; Section 4 fixes notation; Section 5 presents the offline fingerprint and the two online stages; Section 6 describes the prototype; Section 7 reports the RQ1–RQ4 results and the long-tail trade; Section 8 concludes.

2 Background and Motivation

This section establishes what makes SNN serving on CIM clusters distinct from DNN serving, and why two natural design points — capacity-based scheduling and per-spike optimisation — fail. The argument frames the design constraint that drives Section 5: the scheduler needs a workload abstraction that is statistical, small, and consultable in $O(1)$.

2.1 SNN Execution on CIM Clusters

A CIM card hosts a 2D mesh of cores; each core stores a fixed-size slice of synaptic weights in-memory and routes spikes to its neighbours over an on-chip network. This memory-co-located organisation is shared across the production neuromorphic substrates we target — Loihi/Loihi-2 [1, 13], TrueNorth [10], SpiNNaker-2 [9], and Darwin3 [8] — and dictates three properties that shape the scheduling problem.

(i) Bound state. A neural *population* is mapped onto one or more cores at compile time. When any pre-synaptic neuron of population j fires, the post-synaptic membrane updates run locally and out-going spikes are injected into the NoC towards population j's downstream targets. Once the populations of a task are pinned, the task cannot migrate mid-inference without a costly state copy [17], so admission-time decisions are largely irrevocable.

(ii) NoC-dominated contention. The cluster aggregates M such cards; an inter-card interconnect is treated as a separate bandwidth pool. The dominant shared resource is the per-card NoC: when instantaneous injection exceeds its bandwidth ceiling, downstream spikes are queued and tail latency grows.

(iii) Capacity is the wrong yardstick. A card's free-core count is a poor proxy for its real load, since spike injection is sparse and time-varying. Two cards with identical free-core counts can differ by an order of magnitude in their actual NoC pressure once their resident tasks burst together.

2.2 Why Capacity-Based Scheduling Fails

Schedulers from the DNN serving literature place tasks based on scalar resource demands: best-fit or first-fit on heterogeneous resource vectors [14], dominant-resource fairness on a weighted vector [3], power-of-two-choices on instantaneous load [11]. All three are blind to the time dimension. Two tasks that look identical to DRF can have completely different traffic shapes; if the scheduler co-locates a steady-flat tenant and a bursty tenant whose peaks happen to align, the card saturates even though its mean utilisation is well below capacity. We confirm this empirically in Section 7: under bursty arrivals on a 16-card cluster, capacity-based baselines develop a worst-to-best card load ratio above 20×, while their mean utilisation stays under 70 %. Migration-based fixes proposed for GPU clusters [16, 17] can rebalance after the fact, but on CIM hardware the population state is large and the migration cost dwarfs the imbalance window in which the move would have helped.

2.3 Why Schedule-Every-Spike Fails

The opposite extreme — treating each spike as a scheduling decision — is computationally infeasible. Even at modest activity levels (a thousand spikes per task per tick, hundreds of co-resident tasks), a per-spike formulation produces a time-dependent MINLP whose state space exceeds the per-task admission budget by several orders of magnitude. Equally fatal, the scheduler does not know which spikes will fire at admission: the firing pattern depends on the input. A heuristic that schedules at this granularity must either delay admission until the firing trace is observed (eliminating any chance of proactive placement) or guess (eliminating the precision that motivated the per-spike view in the first place).

2.4 The Missing Middle

The scheduler we need lives between these extremes: it must reason about each task's spike activity as a *distribution* that is known at admission time, small enough to evaluate in $O(1)$, and rich enough to expose both spatial topology (which cards' fragmentation matches the task) and temporal shape (where in the card's forecast a new task fits). The next two sections describe how STPS realises this middle: Section 5.2 reduces an SNN's DTDG to four such quantities, and Sections 5.3.1–5.3.2 consume them in closed form.

3 Related Work

Cluster scheduling for DNN serving. Most production cluster schedulers approach placement as bin-packing under multi-dimensional capacity constraints. Best-fit and first-fit on heterogeneous resource vectors [7, 14], dominant-resource fairness [3], and multi-resource generalisations [4] pick a card whose free vector dominates the task's request. These policies are blind to temporal load shape and constitute our baselines in Section 7. Schedulers specialised for DNN training (Gandiva [17], Synergy [11], heterogeneity-aware placement [12]) exploit job-level signals — iteration time, GPU sharing, accelerator heterogeneity — but their unit of work is a deterministic GPU kernel whose footprint does not vary with the input. Predictable DNN serving systems such as Clockwork [5] go further still and make execution time itself a constant by centralising scheduling decisions; this is feasible because the underlying kernel cost is data-independent. None of these systems surfaces a per-tick traffic profile to the scheduler, which is the abstraction STPS relies on, because their workload model does not have one.

Cluster-level admission control. A complementary line of work tackles bursty *arrivals* by predicting future load and provisioning ahead. Workload prediction in serving systems [16] estimates request-rate spikes and pre-scales replicas; admission-control systems for cloud quotas [15] reason about queue depths and fairness across tenants. Both predict the *number* of jobs, not the temporal shape of any single job's hardware demand. The phase-shift stage of STPS is closer to packing in the temporal dimension: rather than predicting future arrivals, it consumes a per-task forecast and chooses a start offset that avoids collision with the card's existing schedule. The closest classical analogue is gang scheduling on parallel machines, but gang scheduling coordinates start times *across cards* to enable communication; STPS coordinates start times *on a single card* to spread NoC pressure.

Neuromorphic mapping and validation. Neuromorphic compilers for production substrates (TrueNorth [10], Loihi [1, 13], SpiNNaker-2 [9], Darwin3 [8]) focus on a static problem: how to partition and place a single model's populations onto cores so that on-chip routing meets latency and fan-out constraints. Validation work on SpiNNaker [7] characterises spike-traffic distributions at this single-model granularity. These tools answer "how does one model fit one chip", which is orthogonal to the multi-tenant placement and timing decisions STPS makes at admission. We treat compiled per-model placements as inputs — the per-population centrality and connected-component structure extracted from the DTDG presupposes that this static mapping has already happened — and ask the next-level question of how to dispatch many such pre-compiled models across many cards.

Fingerprint-driven scheduling. Compressing a workload to a small set of summary statistics is a recurring idea in cluster scheduling and load balancing. STPS's contribution is the choice of fingerprint: four quantities that together capture both the spatial topology ($\overline{K}$, $\mathbf{c}^*_{\max}$) and temporal shape ($E^{(t)}$, β) of an SNN's spike traffic, in a representation small enough to drive admission-time decisions and physical enough to derive from a calibration trace — specifically, from SpikingJelly [2] runs of architectures such as Spikformer [18] — without bespoke instrumentation.

4 Preliminaries

This section fixes the notation used by the rest of the paper. Symbols defined here recur in Section 5 (system) and Section 7 (evaluation); we do not re-introduce them later.

4.1 Workload Model

A submitted task τ is a pre-compiled SNN already partitioned by the vendor toolchain into a set $\mathcal{V}_\tau$ of hardware-aligned populations, each population fitting in one CIM core. An *inference* runs for T discrete ticks. At each tick t, only a subset of populations fires; the resulting per-tick weighted graph $G_\tau^{(t)} = (\mathcal{V}_\tau, \mathcal{E}_\tau^{(t)}, W_\tau^{(t)})$ records the active inter-population edges and the expected spike traffic $W_\tau^{(t)}[i,j]$ flowing from population

i to population j. The sequence $\mathcal{G}_\tau = \{G_\tau^{(1)}, \ldots, G_\tau^{(T)}\}$ is a Discrete-Time Dynamic Graph (DTDG); expectations are taken over a calibration set drawn from the model's evaluation distribution, so $\mathcal{G}_\tau$ depends on the model, not on any specific input.

We never use $\mathcal{G}_\tau$ directly at scheduling time. Instead, Section 5.2 reduces it to four *fingerprint* quantities: the traffic timeline $E_\tau^{(t)} \in \mathbb{R}^T$, global burstiness β_τ, mean active components $\overline{K}_\tau$, and per-population hotspot indicator $\mathbf{c}^*_{\max,\tau} \in \mathbb{R}^{|\mathcal{V}_\tau|}$. The scheduler reads only these.

4.2 Cluster Model

The cluster is a set of M homogeneous cards indexed by $m \in \{1, \ldots, M\}$. Each card holds C CIM cores arranged in a 2D mesh and a NoC with per-card injection ceiling $BW_{\max}$ (spikes/tick). For card m we track its free-core occupancy and, derived from it, the largest contiguous free-block ratio $\rho_m \in [0,1]$ used by Stage 1 dispatch. The card also maintains a per-tick injection forecast $E_m^{(t)} \in \mathbb{R}^H$ over a sliding horizon H, accumulated from all currently-scheduled tasks' offsetted timelines. When the instantaneous demand on card m exceeds $BW_{\max}$, the excess is queued in a pending-traffic buffer rather than dropped; the engine drains it at the cap rate.

Inter-card transfers are modelled as a separate resource pool but never used by our scheduler, which enforces single-card placement: each task lives entirely on one card for the duration of its inference.

4.3 Scheduling Problem

Tasks arrive online according to an arrival process. On each arrival of τ, the scheduler must (a) commit a card $m^*(\tau) \in \{1, \ldots, M\}$ for placement and (b) commit a start offset $\Delta t^*(\tau) \in \{0, 1, \ldots, D_{\max}\}$ after which τ's timeline begins injecting into $E_{m^*}^{(t)}$. Both decisions are irrevocable for the duration of the inference: re-placement requires copying the population state and is prohibitive on CIM hardware.

A scheduler is admissible if, for every τ, it returns $(m^*, \Delta t^*)$ within an $O(1)$ per-task budget that does not scan the underlying DTDG. The system-level objective is to minimise NoC contention — specifically, the fraction of demand pushed back by $BW_{\max}$ and the mean wait in the pending queue — subject to maintaining cluster throughput and bounded tail load imbalance across cards. Section 5 describes the policy STPS uses to make these two decisions; Section 7 measures it against the objectives above.

5 System Design

5.1 Overview

Scheduling SNN inference on a multi-card CIM cluster is hard because spike traffic is both *input-dependent* and *time-varying*: the same model exhibits different injection patterns across inputs and across ticks within one inference. A formulation that schedules every spike exactly is a time-dependent MINLP and is intractable at admission time; reactive runtime mitigation observes congestion only after the NoC has already saturated. STPS resolves this tension by separating what can be known offline from what must be decided online.

Framework. Figure 1 shows the two-phase pipeline. *Offline*, a profiler runs once per model: it traces the SNN over a calibration set, lifts the trace into a Discrete-Time Dynamic Graph (DTDG), and reduces the DTDG to a four-quantity *fingerprint* stored as a small memory-mapped artefact. *Online*, on each task arrival the scheduler reads only the fingerprint and runs two stages in sequence: macro-card dispatch picks a card m^*, then phase-shift picks a start offset Δt^*. The pair $(m^*, \Delta t^*)$ is committed to the chosen card's forecast and returned to the runtime. Both stages run in $O(1)$ of the model size (linear in $D_{\max} \cdot H$ for Stage 2), so admission cost does not grow with model complexity.

Stage 1 — Macro-Card Dispatch. The first stage chooses *where* the task lands. For each card m with sufficient free capacity, it computes a single scalar score $\mathcal{S}_m(\tau)$ that combines two pressures: (i) how well the task's topological cohesion $\overline{K}_\tau$ matches the card's largest contiguous free-block ratio ρ_m, and (ii) whether the card already hosts bursty workloads when τ is itself bursty ($\beta_\tau > \beta_{\text{hi}}$). The card minimising $\mathcal{S}_m$ wins. The score is a weighted sum rather than a lexicographic rule, so it degrades smoothly between empty and saturated regimes without policy switching.

Stage 2 — Temporal Phase-Shifting. The second stage chooses *when* the task starts. Spatial dispatch alone does not prevent two bursty tasks from peaking on the same tick; Stage 2 exploits the fact that the task's traffic timeline $E_\tau^{(t)}$ is known at admission. It scans the $D_{\max}+1$ candidate offsets $\Delta t \in \{0, \ldots, D_{\max}\}$ and commits the one whose superposition $E_{m^*}^{(t)} + E_\tau^{(t-\Delta t)}$ has the smallest peak under the bandwidth ceiling $BW_{\max}$. The chosen offset is added back to $E_{m^*}^{(t)}$, updating the forecast that the next admission will see.

The rest of this section justifies the fingerprint design (Section 5.2) and details each online stage (Sections 5.3.1 and 5.3.2); the hotspot indicator and discussion follow in Sections 5.4 and 5.5.

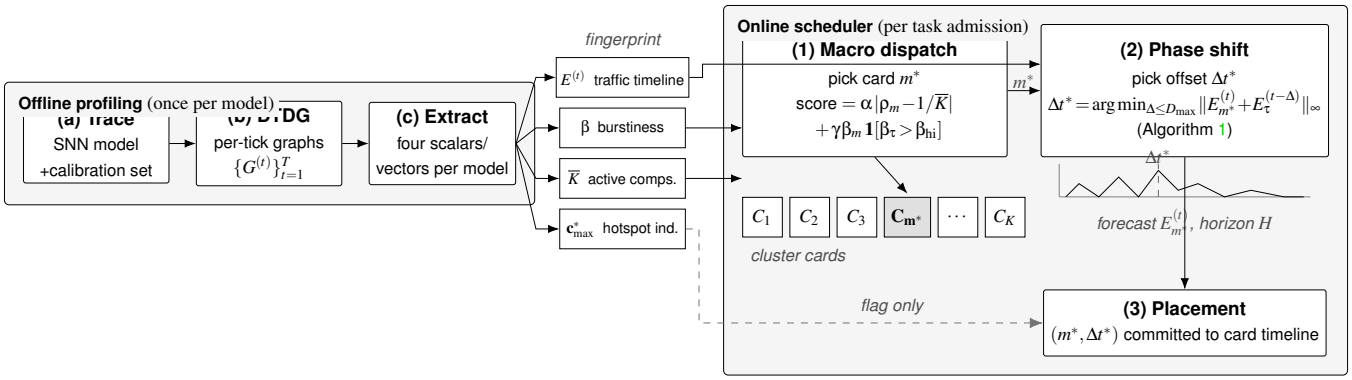


Figure 1: STPS pipeline. *Offline:* a model is traced, lifted into a discrete-time dynamic graph, and reduced to four fingerprints. *Online:* each admission runs in two stages — macro-dispatch picks a card m^* from $\{C_1, \ldots, C_K\}$ using $\overline{K}$ and β, then phase-shift picks a start offset $\Delta t^* \le D_{\max}$ that fits the task's traffic timeline into a valley of the card's forecast. The hotspot indicator $\mathbf{c}^*_{\max}$ is recorded but does not drive placement in the current system (dashed arrow).

5.2 Workload Fingerprint

We model an SNN inference as a Discrete-Time Dynamic Graph (DTDG) $\mathcal{G} = \{G^{(1)}, \ldots, G^{(T)}\}$, where each snapshot $G^{(t)} = (\mathcal{V}, \mathcal{E}^{(t)}, W^{(t)})$ shares the same vertex set $\mathcal{V}$ of hardware-aligned neural populations (each population fits within one CIM core) and a per-tick weight tensor $W^{(t)}_{ij}$ measuring the expected spike traffic on edge $i \to j$. The expectation is taken over a calibration set drawn from the model's evaluation distribution, so the fingerprint is independent of any single input.

From this trace we extract three quantities that the online scheduler consumes:

- **Traffic timeline** $E^{(t)} \in \mathbb{R}^T$**.** The total spike traffic at tick t, averaged over the calibration set. $E^{(t)}$ is the only fingerprint that retains temporal shape.
- **Global burstiness** $\beta = \max_t E^{(t)} / \overline{E}$**.** Peak-to-mean ratio of the traffic timeline; $\beta \approx 1$ for steady workloads, $\beta \gg 1$ for event-driven bursts.
- **Mean active components** $\overline{K}$**.** Mean number of connected components in the active-edge subgraph of $G^{(t)}$, averaged over t. $\overline{K} \to 1$ indicates a topologically cohesive model; $\overline{K} \gg 1$ indicates several concurrent, weakly-coupled subflows.

A fourth quantity, the per-population maximum in-eigenvector centrality $\mathbf{c}^*_{\max}[i] = \max_t c^{(t)}_i$, is recorded as a hotspot indicator and discussed in Section 5.4.

The fingerprint is on the order of tens of kilobytes per model and is computed once at deployment time; subsequent scheduling decisions read it as a memory-mapped artefact.

5.3 Online Scheduler

This subsection details the two stages summarised in Section 5.1.

5.3.1 Stage 1: Macro-Card Dispatch

The cluster constraint we enforce is single-card placement: each model resides entirely on one card. The dispatch problem therefore reduces to selecting one card among those with sufficient free cores and memory for τ. We score each feasible card m by

$$\mathcal{S}_m(\tau) \;=\; \alpha \left| \rho_m - \tfrac{1}{\overline{K}_\tau} \right| \;+\; \gamma \beta_m \cdot \mathbb{1}[\beta_\tau > \beta_{\text{hi}}], \qquad (1)$$

where ρ_m is the largest contiguous free-block ratio on card m and β_m is its running mean burstiness. The first term matches the task's topological cohesion against the card's available shape: cohesive tasks ($\overline{K}_\tau \to 1$) are drawn to cards with large contiguous blocks ($\rho_m \to 1$), while decoupled tasks ($\overline{K}_\tau \gg 1$) absorb fragmented cards without penalty. The second term keeps high-burstiness tasks away from cards that already host bursty workloads; it activates only above a threshold β_{hi} so that flat traffic incurs no temporal penalty. The card with minimum $\mathcal{S}_m$ wins.

5.3.2 Stage 2: Temporal Phase-Shifting

Cards co-locate multiple tasks, and their traffic timelines superpose on the same on-chip network. Spatial dispatch alone does not prevent two bursty tasks from peaking on the same tick. Stage 2 exploits the fact that the timeline $E^{(t)}_\tau$ is known at admission to delay τ's injection so that its peaks fall into the valleys of the existing card-level forecast.

Let $E^{(t)}_m$ be the forecasted background traffic on the selected card m and $D_{\max}$ the deadline-bounded shift budget.

Algorithm 1 Phase-Shift Selection

Require: Background E_m, task E_τ, budget $D_{\max}$, ceiling $BW_{\max}$
Ensure: Offset $\Delta t^* \in [0, D_{\max}]$ (offset 0 if none fits under $BW_{\max}$)

```
p* ← ∞, Δt* ← 0                          ▷ default: admit unshifted
for Δt = 0 to D_max do
    p ← max_t ( E_m^(t) + Shift(E_τ, Δt)^(t) )
    if p ≤ BW_max and p < p* then
        p* ← p, Δt* ← Δt
    end if
end for
E_m ← E_m + Shift(E_τ, Δt*)
return Δt*
```

The optimal offset is

$$\Delta t^* \;=\; \arg\min_{\Delta t \in [0, D_{\max}]} \; \max_{t} \left(E_m^{(t)} + E_\tau^{(t-\Delta t)} \right), \qquad (2)$$

subject to the resulting peak staying below the per-card bandwidth ceiling $BW_{\max}$. When no offset in $[0, D_{\max}]$ satisfies the ceiling, the task is admitted unshifted (offset 0) — it is never dropped — and simply contributes its share to congestion; we call this an *offset-infeasible* admission, and it is rare except at extreme load. Algorithm 1 expresses this as a single linear scan over $D_{\max}+1$ candidate offsets; the cost is $O(D_{\max} \cdot T)$ per task.

5.4 Spatial Mapping and Hotspot Indicator

Once a card and offset are fixed, populations are mapped to PIM cores by the vendor compiler under standard hardware constraints. The fingerprint exposes $\mathbf{c}^*_{\max}$ as a per-population indicator of compute concentration: a population whose centrality exceeds a threshold θ_c is a candidate for splitting across adjacent cores, trading a small spatial footprint for reduced peak SOP pressure. In the current system this flag is recorded on the task but does not drive placement; we leave the centrality-driven splitting pass, and its end-to-end evaluation, to future work.

5.5 Discussion

Two design choices warrant comment. First, all four fingerprint quantities are scalars or short vectors; the scheduler never inspects the underlying $T \times |\mathcal{V}|^2$ trace at runtime. This keeps admission cost independent of model size and is what allows STPS to interleave with arrival rates of thousands of tasks per second. Second, the macro-dispatch score in Equation 1 is deliberately a weighted sum rather than a lexicographic policy: the same scalar handles both empty and saturated regimes, so STPS degrades gracefully as cluster load increases rather than switching between policies.

6 Implementation

We implement STPS as a Python prototype on top of a discrete-event simulator of a multi-card CIM cluster. The system is partitioned into three packages so that the offline profiler, the online scheduler, and the simulation harness can evolve independently. The fingerprint extractor (FINGERPRINT/, ∼2 kLoC) lifts an SpikingJelly [2] run into a DTDG and reduces it to the four quantities of Section 5.2; the resulting artefact is persisted as a memory-mapped `.npz` blob (∼tens of kilobytes per model) and re-loaded lazily at admission time, so cold-start cost is a single file read. The online scheduler (SCHEDULE/, ∼0.8 kLoC) implements both stages and a pluggable registry that exposes RR, BestFit, DRF, P2C, and the STPS variants behind a uniform interface; the registry also hosts the stage ablations as named entry points so the evaluation harness can sweep policies without touching the simulator core.

The simulation harness (SIMULATION/, ∼0.5 kLoC) implements the two-tier tick loop — per-card placement, NoC bandwidth accounting, pending-traffic queue — in pure NumPy; we deliberately avoid a hardware-cycle-accurate model because the scheduling decision is made entirely at admission, and the per-tick simulation only has to be accurate *enough* to faithfully expose the bandwidth ceiling and queue dynamics. The whole system is around 4.6 kLoC of Python and 1.8 kLoC of pytest tests, exercised through a `Makefile` that drives both the per-policy runs and the cross-scheduler comparison sweeps used in Section 7. The fingerprint extractor can also be invoked standalone (`python -m fingerprint.cli`) to produce synthetic fingerprints, which is what the steady-flat and pulse-train workloads in Section 7.1 are built from.

7 Evaluation

The evaluation is organised around four questions, each closing one line a reviewer would otherwise leave open. **RQ1** (Section 7.2): does STPS improve end-to-end NoC behaviour on a realistic mixed-fingerprint workload, and at what cost? **RQ2** (Section 7.3): which stage contributes what — can the gain be attributed to spatial dispatch, to phase-shift, or only to their combination? **RQ3** (Section 7.4): when does STPS help and when does it hurt, as we sweep the workload's predictability, the bandwidth cap, and the offset budget across their operational ranges? **RQ4** (Section 7.5): what tail cost does STPS pay, and is it the right trade for a CIM cluster? The short answers: STPS is the only scheduler that lowers NoC congestion in every cell tested at a 1–2 % throughput cost; the two stages contribute along complementary axes and neither alone suffices; STPS helps where topology is predictable and the cap binds, and degrades gracefully elsewhere; and the entire cost is concentrated in the worst ∼1 % of per-task delays.

7.1 Experimental Setup

Simulator. We use a discrete-event, trace-driven simulator of a multi-card CIM cluster (Section 5). Each card holds 256 cores wired by a 2D mesh NoC. The engine accounts for per-card instantaneous demand, NoC bandwidth cap $BW_{\max}$, and a pending-traffic queue that absorbs over-budget injection rather than dropping the task; the queue is drained at the cap rate so no task is ever rejected and the cluster's completion rate is always 1.0. All schedulers share the simulator and the same per-card placement primitive; they differ only in the admission-time dispatch and (for STPS) phase-shift logic.

Workloads. Each task draws a fingerprint from a *mixed* set: four synthetic templates (steady-flat, two pulse trains with periods $T = 8$ and $T = 16$, and a heavy-tail burst) plus three real SpikingJelly [2] traces (Spikformer [18]-CIFAR10, QKFormer-CIFAR10, SpikingResFormer-Tiny-ImageNet). Task arrivals follow either a Poisson process, a bursty arrival pattern that bunches admissions into onset windows, or a 1:1 mix of the two.

Fidelity. The simulator abstracts the cluster at the granularity our claims live at: per-card, per-tick NoC injection. Three properties make this sufficient to support the conclusions. First, the traffic timeline $E_\tau^{(t)}$ that drives every placement and phase-shift decision is *trace-derived*, not parametric: the three real fingerprints are extracted from actual SpikingJelly spike tensors via the offline DTDG pipeline (Section 5.2), so the burst shapes the scheduler reacts to are the ones the models actually produce. The four synthetic templates exist only to sweep extremes (perfectly flat, single-period, heavy-tail) that the real traces sample but do not isolate. Second, the contended resource we model — the NoC bandwidth cap and the pending-traffic queue draining at that cap — is the resource SNN inference on CIM actually serialises on: with weights resident in memory, inference is communication-not compute-bound, so an injection-rate model captures the first-order bottleneck while remaining agnostic to per-core compute timing that scheduling does not influence. Third, the cap itself is not a tuned knob: $BW_{\max}$ is fixed at the p75 of the per-card demand observed under an *uncapped* run, a workload-derived operating point at which every scheduler — ours included — sees congestion, rather than a value chosen to flatter STPS. We report the sensitivity of our conclusions to this choice across three cap-binding regimes in Section 7.4.

Configuration. Unless stated otherwise the cluster has $M = 4$ cards, runs for 512 ticks, $BW_{\max} = 9 \times 10^5$ (the p75 of the per-card demand observed under an uncapped run, so all schedulers see meaningful congestion), $D_{\max} = 2$, $H = 64$. We average each configuration over 5 seeds (21, 42, 63, 84, 105) and report ±95 % confidence half-widths where space permits; steady-state metrics skip the first and last 64 ticks of each run. The 16-card configuration scales the task count to 2048 to keep utilisation comparable.

Baselines. Round-robin (**RR**), best-fit on free cores (**BestFit**), dominant-resource fairness (**DRF**) [3], and power-of-two-choices (**P2C**) [11]. All four are spatial-only: they ignore fingerprints and inject tasks with zero offset. For the Stage-A ablation we isolate STPS's macro-dispatch as the spatial-only variant STPS-SPATIAL; for the Stage-B ablation we additionally pair each baseline with the phase-shift module (+PS). The full STPS scheduler is the STPS-SPATIAL+PS pairing.

Metrics. We report (i) cross-card load coefficient of variation *card-CV*, Jain's fairness index *JFI* [6] (JFI $= (\sum_m L_m)^2 / (M \sum_m L_m^2) \in (1/M, 1]$, 1 = perfectly balanced), the load-imbalance factor *LIF* (LIF $= \max_m L_m / \overline{L} \geq 1$, the busiest card's load relative to the mean), and the max/min load ratio — four complementary views of across-card balance, with card-CV/JFI capturing dispersion and LIF/max-min the tail outlier; (ii) cluster *throughput* in tasks/tick, (iii) NoC *congestion ratio* (the fraction of demand pushed back by the cap) and the mean wait in the pending queue (*cong-wait*, ticks), and (iv) the p99 of per-task end-to-end delay, which we use to characterise the tail trade in Section 7.5. To separate Stage-B's intrinsic offset from genuine queueing, we additionally report the *cold-start-excluded throughput*, $N_{\text{completed}} / (T_{\text{steps}} - \overline{T_{\text{cold}}})$, where $\overline{T_{\text{cold}}}$ is the per-run mean start offset committed by phase-shift.

7.2 RQ1: Congestion-Bounded End-to-end Behaviour

The headline. Table 1 reports STPS against the four baselines on the same workload at two cluster sizes. The headline is the congestion column: STPS wins every arrival/scale cell on both the congestion ratio and the mean queueing wait. No baseline wins on either metric in any cell. The size of the win is small in absolute terms (5–8 % on the ratio, 7–9 % on the wait) but its consistency is what distinguishes STPS from the baselines, which trade leadership across cells.

The cost. STPS does not dominate the spatial metrics. Across the cells tested card-CV is 2–7 % higher than the best baseline and throughput is 1–2 % lower. Two distinct mechanisms produce this gap. First, Stage B injects a non-zero start offset whenever the forecast peak would collide with the card's existing schedule; this offset appears as the cold-start budget $\overline{T_{\text{cold}}} \approx 1.0$ tick under $D_{\max} = 2$ and accounts for ∼0.2 % of the throughput delta (excluding it from the throughput denominator closes only 0.16–0.17 % of the 1–2 % gap). The remaining 0.8–1.8 % is contention avoidance: when phase-shift defers a task into a residual valley, the card injects strictly less per tick. The cluster trades a fraction of average throughput for a guaranteed reduction in instantaneous congestion. Second, STPS chooses cards by topology fit rather than by lowest-load match, so card-CV — which measures across-card variance of instantaneous served load — can be slightly higher than a pure load-balancer's, even when the worst-card tail is smaller (see below).

Table 1: End-to-end comparison on the mixed-fingerprint workload (5 seeds, mean values; bold = best in column). STPS is the *only* scheduler that wins on both congestion metrics in every {arrival, scale} cell tested at a 1–2 % throughput cost. The advantage in the worst-card max/min ratio sharpens with scale: at 16 cards STPS holds it under 13.5× while every baseline develops a 14–24× skew.

Scale	Arrival	Scheduler	card-CV ↓	JFI ↑	max/min ↓	throughput ↑	cong. ratio ↓	cong. wait ↓
4 cards	Poisson	RR	**0.267**	**0.911**	7.42	1.318	0.269	9.26
		BestFit	0.288	0.910	7.95	1.303	0.260	9.06
		DRF	0.288	0.910	7.95	1.303	0.260	9.06
		P2C	0.275	0.910	6.84	1.314	0.265	9.13
		STPS	0.293	0.906	**6.61**	1.292	**0.247**	**8.42**
	Bursty	RR	**0.287**	**0.906**	6.11	1.293	0.258	8.87
		BestFit	0.286	0.904	6.06	**1.296**	0.258	8.89
		DRF	0.286	0.904	6.06	**1.296**	0.258	8.89
		P2C	0.294	0.901	**5.80**	1.286	0.255	8.86
		STPS	0.307	0.896	6.23	1.269	**0.239**	**8.14**
	Mixed	RR	**0.265**	**0.912**	7.13	**1.325**	0.267	9.31
		BestFit	0.292	0.910	7.36	1.309	0.259	9.21
		DRF	0.292	0.910	7.36	1.309	0.259	9.21
		P2C	0.275	0.910	**6.79**	1.316	0.263	9.25
		STPS	0.300	0.906	6.92	1.297	**0.245**	**8.56**
16 cards	Poisson	RR	**0.305**	**0.912**	16.90	5.346	0.275	9.45
		BestFit	0.307	0.911	21.52	5.321	0.274	9.41
		DRF	0.307	0.911	21.52	5.321	0.274	9.41
		P2C	0.303	0.913	21.78	5.339	0.276	9.53
		STPS	0.317	0.906	**13.42**	5.274	**0.256**	**8.61**
	Bursty	RR	**0.313**	**0.907**	21.09	5.263	0.268	9.14
		BestFit	0.318	0.904	14.62	5.251	0.267	9.12
		DRF	0.318	0.904	14.62	5.251	0.267	9.12
		P2C	0.311	0.908	23.90	**5.279**	0.271	9.25
		STPS	0.325	0.899	**12.09**	5.212	**0.252**	**8.45**

The scale signal. The most consequential result in Table 1 is the max/min column at 16 cards. Capacity-based baselines develop a tail-card skew of 14–24× at the scale where the cluster has the most placement choices to make; STPS holds it at 12.1 (bursty) and 13.4 (Poisson). The advantage *widens* with scale: at 4 cards the gap to the best baseline is 0.1–0.6×, at 16 cards it is 2.5–11.7×. This is consistent with the design intent of Section 5.3.1: the macro-dispatch score becomes more selective as more cards are available to score against, so the topology signal that fingerprint-aware dispatch carries scales with cluster size in a way that capacity-only policies cannot match.

Is the trade favourable? The RQ1 picture is therefore not "STPS is a better balancer", but "STPS is the only scheduler whose objective is the right objective for a CIM cluster". In a substrate where the NoC is the dominant shared resource (Section 2.1), spending 1–2 % of throughput to consistently reduce NoC pressure — and, at scale, to keep the worst card under 13.5× instead of letting it run at 24× the cluster mean — is the trade an operator should want.

Table 2: Throughput decomposition at 16 cards: cold-start-excluded throughput closes only 0.16–0.17 % of the 0.6–1.4 % gap to the best baseline. The remaining gap is contention-avoidance — the price STPS pays to keep the NoC under its cap.

Arrival	Scheduler	$\overline{T_{\text{cold}}}$	tput	$\text{tput}_{\text{excl. cs}}$
Poisson	best baseline (RR)	0.00	5.346	5.346
	STPS	1.02	5.274	5.282
Bursty	best baseline (P2C)	0.00	5.279	5.279
	STPS	1.01	5.212	5.221

7.3 RQ2: Which Stage Contributes What

STPS combines two stages, and a reviewer is entitled to ask which one earns the result. We answer with a single ablation table that holds the workload and cap fixed ($BW_{\max} = 9 \times 10^5$, $D_{\max} = 2$) and reads off three rows per policy: the spatial baseline alone (BASE), the same baseline with phase-shift bolted on (+PS), and STPS-SPATIAL — our macro-dispatch

in isolation — with and without phase-shift. The bottom row, STPS-SPATIAL+PS, is the full scheduler. Reading the table down each policy isolates Stage B's marginal effect; reading across the BASE rows isolates Stage A's. Stage B is policy-agnostic by construction: it consumes the traffic timeline $E^{(t)}$ and the card forecast, scans up to $D_{\max}$ offsets, and commits the one whose forecast peak is lowest under the cap (Algorithm 1), so it can be attached to any spatial policy.

Phase-shift collapses congestion at $D_{\max}=2$. Table 3 reports the ablation. The dominant result is a single number per row: attaching phase-shift to RR, BestFit, DRF, or P2C drives the NoC congestion ratio from $\sim$0.26 to below 0.01 — a 26× to 200× reduction — and the mean queue wait from 8.9 ticks to under one tick. The mechanism is direct: by deferring each task's onset by at most $D_{\max}=2$ ticks the optimiser finds a slot whose superposition on the card forecast stays below $BW_{\max}$. The cost is small: mean offsets stay at $\sim$1 tick, p99 grows by 2 ticks, and throughput falls by 6–18 %. STPS-SPATIAL+PS (full STPS) sees a smaller absolute congestion drop ($0.26 \rightarrow 0.16$) because Stage A has already steered topology-cohesive and bursty tasks toward cards whose forecasts have less easy slack to absorb; the slack phase-shift consumes for the baselines is gone.

The card-CV anomaly is a feature, not a regression. A striking secondary observation in Table 3 is that card-CV *rises* when phase-shift is attached to any spatial baseline (e.g., P2C's $0.275 \rightarrow 0.532$). The mechanism is mechanical: card-CV measures the across-card variance of instantaneous served load; once phase-shift de-synchronises tasks that previously peaked together, each card's per-tick demand becomes spikier even as the cluster-wide queue empties. The metric the operator cares about — congestion ratio and cong-wait — moves the right way, but card-CV registers the de-synchronisation as a regression. This is why STPS combines both stages: Stage A flattens the across-card distribution at admission time so that Stage B's de-synchronisation does not blow card-CV up, while Stage B flattens each card's timeline so that Stage A's topology choices do not stack peaks. The mid-table rows of Table 3 make this concrete: full STPS lands at card-CV 0.293/0.307 vs the baseline-plus-phase rows' 0.31–0.60.

Three workpoints, one mechanism. Phase-shift's effect on every other metric flips when we change a single dimension: how hard the bandwidth cap binds. To isolate this, we re-run the full Stage-B ablation under three operating regimes that vary independently along the cap-binding axis while keeping every other knob fixed (4 cards, mixed fingerprints, $H=64$, $D_{\max}=16$, 5 seeds; $D_{\max}=16$ amplifies the signal that $D_{\max}=2$ shows in attenuated form). Regime A (binding, $BW_{\max}=9\times10^5$, 800 tasks) places demand at the p75 of an uncapped run, so the cap clips roughly 26 % of the demanded injection and forces a deep pending-traffic queue. Regime B (non-binding, $BW_{\max}=5\times10^6$, 800 tasks) raises the cap by 5.5× so demand never reaches it; baselines never queue. Regime C (lightly binding, $BW_{\max}=9\times10^5$, 400 tasks) halves the task count, leaving the same cap but only $\sim$9 % baseline congestion. The three regimes are designed to span the operationally relevant range: a CIM cluster that is over-provisioned (B), correctly provisioned (C), or under-provisioned (A).

The decomposition, in one line. Neither stage alone is the scheduler. Phase-shift on a capacity baseline zeroes congestion but inflates card-CV (the +PS rows); macro-dispatch alone (STPS-SPATIAL) holds card-CV near the baseline but leaves congestion at $\sim$0.26; only the pairing reaches both low congestion (0.16) and a controlled card-CV (0.293/0.307). Stage A buys the across-card distribution, Stage B buys the per-card timeline, and the cross term — not stacking peaks that the other stage just flattened — is why the combination beats the sum of the ablations.

7.4 RQ3: When STPS Helps and When It Hurts

STPS is not uniformly best, and a paper that claimed so would be wrong. This section maps the boundary explicitly along three axes an operator controls: how predictable the workload's topology is (does Stage A's score carry signal?), how hard the bandwidth cap binds (is there congestion for Stage B to lever?), and how large the offset budget $D_{\max}$ is (how much tail may phase-shift spend?). The headline is that STPS's two stages are *conditional* levers — each helps in the regime it was designed for and degrades gracefully outside it — and that the conditions are observable before deployment.

Predictability: when Stage A's score carries signal. Stage A consumes only two fingerprint quantities — the topological cohesion $\overline{K}_\tau$ and the burstiness β_τ — and emits a single scalar score. The question is: when does that score buy anything? We answer with two sweeps. The first varies cluster *utilisation* on a single workload mix; the second varies the *fingerprint composition* of the workload at fixed utilisation. Both are run on a 4-card cluster with 512 cores per card so that Stage A is exercised in isolation (without $BW_{\max}$ binding and without Stage B influence). All numbers below are 5-seed means; we report ±95 % half-widths where they affect interpretation.

Utilisation determines whether topology matters. Table 4 reveals a clean regime structure. Below 70 % utilisation BestFit and DRF win because there is enough free capacity that almost any placement is correct, and the $\overline{K}_\tau$-aware score adds noise rather than signal. Between 0.85 and 0.95 the four

Table 3: Stage decomposition: each spatial policy (BASE) paired with phase-shift (+PS), and STPS's own macro-dispatch STPS-SPATIAL with phase-shift added; the full scheduler is STPS-SPATIAL+PS. 4 cards, mixed fingerprints, $BW_{\max} = 9 \times 10^5$, $D_{\max} = 2$, 5 seeds. Phase-shift collapses NoC congestion from $\sim$0.26 to ≤ 0.01 on every spatial baseline; on full STPS the drop is $0.26 \rightarrow 0.16$ because Stage A's topology- and burstiness-aware placement has already absorbed the easiest temporal slack.

Arrival	Policy	card-CV	throughput	cong. ratio	cong. wait	p99 delay	mean offset
Poisson	RR BASE	**0.267**	1.318	0.269	9.26	**15.0**	0.00
	RR +PS	0.313	1.094	**0.000**	**0.00**	17.0	1.05
	BestFit BASE	0.288	1.303	0.260	9.06	15.0	0.00
	BestFit +PS	0.321	1.149	0.000	0.00	17.0	0.98
	P2C BASE	0.275	1.314	0.265	9.13	15.0	0.00
	P2C +PS	0.532	1.079	0.010	0.40	17.0	1.03
	STPS-SPATIAL	0.268	1.292	0.263	9.10	15.0	0.00
	stps-spatial+ps	0.293	1.292	0.160	5.10	17.0	0.99
Bursty	RR BASE	0.287	1.293	0.257	8.87	15.8	0.00
	RR +PS	0.393	1.041	**0.000**	0.30	17.0	0.95
	BestFit BASE	0.286	1.296	0.258	8.89	15.8	0.00
	BestFit +PS	0.408	1.076	0.000	0.15	17.0	0.89
	P2C BASE	0.294	1.286	0.255	8.86	15.8	0.00
	P2C +PS	0.600	1.015	0.000	0.10	17.0	0.98
	STPS-SPATIAL	0.277	1.292	0.260	8.95	15.8	0.00
	stps-spatial+ps	0.307	1.269	0.157	5.05	17.0	0.99

Table 4: Stage A under a utilisation sweep (4 cards $\times$ 512 cores, Poisson, mixed fingerprints, 5 seeds). At low load BestFit/DRF dominate because empty-capacity signals overwhelm topology signals; from 0.85 onward Stage A converges into the leader band and at 0.95 all schedulers cluster in a 0.20–0.23 strip — the regime where physical saturation removes spatial degrees of freedom.

Util.	RR	BestFit	DRF	P2C	STPS-A
0.30	0.722	**0.652**	0.657	0.908	1.004
0.50	0.493	0.402	**0.392**	0.632	0.592
0.70	0.363	**0.292**	0.293	0.407	0.386
0.85	0.257	0.224	**0.223**	0.293	0.249
0.95	0.218	0.202	**0.200**	0.230	0.207

Table 5: Stage A under a fingerprint-composition sweep (4 cards, Poisson, 70 % utilisation). Baselines ignore fingerprints and post the same card-CV everywhere; only Stage A reads the workload's topology and therefore changes. At 100 % steady-flat Stage A delivers the lowest card-CV of any scheduler — the design intent. The advantage erodes as the workload becomes more bursty, because $H = 64$ forecast cannot resolve next-tick peaks.

Flat / bursty	RR	BestFit	DRF	P2C	STPS-A
100 / 0	0.363	0.292	0.293	0.407	**0.290**
75 / 25	0.363	**0.292**	0.293	0.407	0.320
50 / 50	0.363	**0.292**	0.293	0.407	0.340
25 / 75	0.363	**0.292**	0.293	0.407	0.411
0 / 100	0.363	**0.292**	0.293	0.407	0.555

spatial policies converge into a tight 0.20–0.23 band: at saturation no policy has spatial degrees of freedom to exploit and all schedulers are equally helpless. The interesting window is precisely the gap, where Stage A is within 11 % of BestFit and ahead of RR/P2C — the operating range for which fingerprint-aware dispatch was designed.

Fingerprint predictability determines whether Stage A wins. Table 5 fixes utilisation at 70 % and varies the workload from purely steady-flat to purely bursty fingerprints. Baselines do not read fingerprints, so their column is constant at every row. Stage A's column tells the whole story: at 100 % steady-flat it is the best scheduler in the table (card-CV 0.290, -20.1 % vs RR, -0.9 % vs BestFit) — the design intent of [17]-style topology-aware placement realised. As the bursty share grows the advantage erodes and, at 100 % bursty, Stage A becomes the worst of the five. This is not a bug. Stage A scores cards in $O(1)$ from the $H = 64$ forecast horizon, and a $T = 8$ pulse-train fingerprint's next peak can be eight horizons away; capacity-based policies, which ignore the future altogether, become better defaults in the regime where the future is unknowable from the fingerprint alone. This is the first boundary: Stage A is a predictability lever, strongest when the fingerprint resolves the near future and neutral-to-negative when it cannot. It is exactly the bursty regime where Stage A weakens that Stage B, analysed next, takes over — the two stages cover complementary halves of the predictability axis.

Table 6: How each metric responds as the phase-shift budget $D_{\max}$ grows from 2 to 16 (light regime C, 4 cards, mixed fingerprints, 5 seeds). Congestion is a switch, not a knob; load-balance improves with diminishing returns and saturates near $D_{\max}=8$; tail delay and throughput cost grow monotonically with the offset budget. The three columns pull in opposite directions, which is what fixes a small default.

Metric	Response to $D_{\max}\uparrow$	Detail
Cong. ratio (non-STPS)	flat at 0 from $D_{\max}=2$	switch, not knob
Cong. ratio (STPS)	slow $\downarrow$	$0.087\rightarrow0.079$
card-CV / JFI / LIF	$\downarrow$, saturating	knee near $D_{\max}=8$
Max/min	$\downarrow$	8–13 $\rightarrow$ 3–6
p99 / mean offset	$\uparrow$ monotone	offset $0.5\rightarrow4$ ticks
Throughput	partial recovery	stays below baseline

Cap binding: when Stage B has congestion to lever. Phase-shift only relieves congestion where the cap binds. Sweeping the cap across three regimes — binding ($BW_{\max}=9\times10^5$, 800 tasks; $\sim$26 % baseline congestion), non-binding (5×10^6, demand never reaches the cap), and lightly binding (9×10^5, 400 tasks; $\sim$9 %) — the congestion ratio drops to zero wherever there is congestion to remove (binding and light) and is undefined where there is none (non-binding). The load-balance metrics, by contrast, *reverse sign* with the regime: phase-shift lowers card-CV when the cap does not bind but raises it when the cap is already clipping peaks, because a binding cap is itself a crude flattener and de-synchronisation moves cards off that artificial uniformity. This sign-flip is universal — it holds for every baseline and for STPS-SPATIAL alike, so it is a property of the regime, not of STPS. The practical reading: STPS's sweet-spot is a correctly- or over-provisioned cluster; pushed into heavy congestion, no scheduler improves load-balance, and the right response is provisioning, not a smarter policy. Appendix A gives the full per-policy grid.

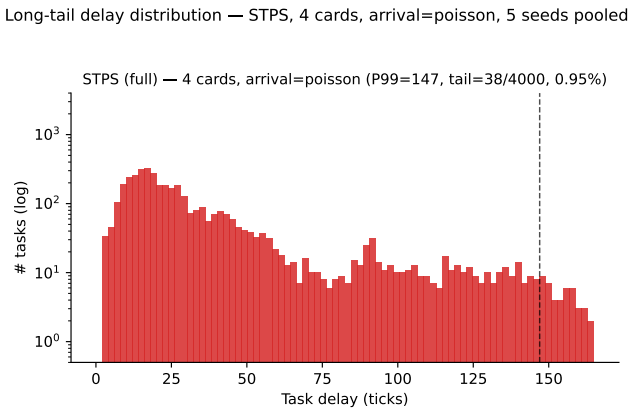
Offset budget: how much tail phase-shift may spend. A budget sweep $D_{\max}\in\{2,4,8,16\}$ at the light cap (Table 6) shows the three groups of metrics pulling against each other. Congestion relief is binary: the moment phase-shift is enabled, every capacity baseline's congestion ratio drops from $\sim$0.09 to zero, and no larger budget improves on zero. Load-balance (card-CV, JFI, LIF) improves with diminishing returns and saturates near $D_{\max}=8$ — the RR card-CV bottoms at 0.44 for $D_{\max}=8$ then rebounds at 16. But tail delay moves the opposite way: the mean injected offset grows from $\sim$0.5 to $\sim$4 ticks across the sweep, dragging p99 up monotonically and clawing back the throughput that the cap rejections cost. The budget that buys the congestion switch and most of the load-balance gain while paying the least tail and throughput is the smallest one. We adopt $D_{\max}=2$ as the default: it already zeroes congestion, recovers the bulk of the max/min compression (8–13$\times$ $\rightarrow$ 3–6$\times$), and holds the mean offset under one tick. On full STPS the case is sharper still — a larger budget yields no congestion or balance gain (Stage A has already consumed the spatial slack) and only inflates p99, so $D_{\max}=2$ is strictly preferred. The three-regime analysis above explains which of these signals would persist or invert under a heavier cap, even at this attenuated budget.

7.5 RQ4: The Tail Cost of Holding Load Flat

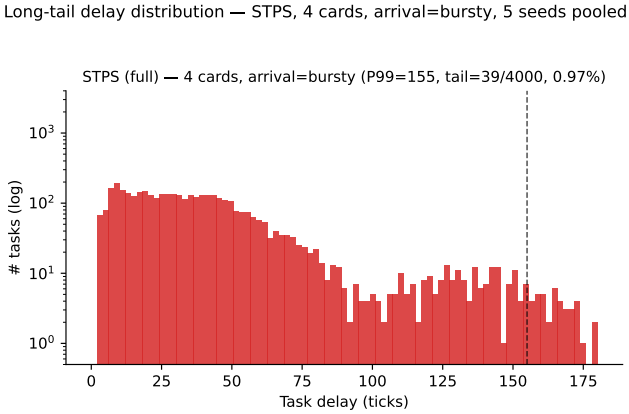
Averages flatter cluster schedulers. A scheduler that improves the median can still fail an SLA at the tail, and the tail is exactly where STPS's trade-off lives. Figure 2 plots $\Pr[\text{delay}>x]$ for both the 4-card and 16-card clusters under both arrival modes. The four panels tell a consistent story: up to roughly the 90th percentile every scheduler tracks every other within a few ticks; the curves only fan out in the upper-right corner. By construction, the tasks that populate that corner are the long-tail tasks — the same bursty, high-β_τ population that defines p99 in the first place — and STPS's slower-decaying curve sits to the right of the baselines precisely on those tasks. At p99 the gap is 6 ticks under Poisson and 12 ticks under bursty arrivals on 16 cards, and the 4-card panels show the same shape at smaller absolute scale. Crucially, the curves are indistinguishable on the body: STPS does not slow median or p90 tasks; it spends its delay budget exclusively on the worst decile.

Where the tail comes from. Phase-shift commits each task to a non-zero start offset whenever the chosen card's forecast peak would otherwise collide with the task's burst. Over a long run, the offset accumulates on the bursty tasks — the ones whose β_τ is large — because they are the ones whose peaks need shifting. The same tasks dominate the worst-card load in the baselines; STPS finishes them 5–10 % later and uses the slack to keep the worst card under control. In exchange, the cluster as a whole sees the gains already documented: congestion ratio $-$5–8 %, mean queue wait $-$7–9 %, 16-card max/min 14–24$\times$ $\rightarrow$ 12$\times$. The same offsets that lengthen the worst 1 % of tasks are what flatten the cards underneath them.

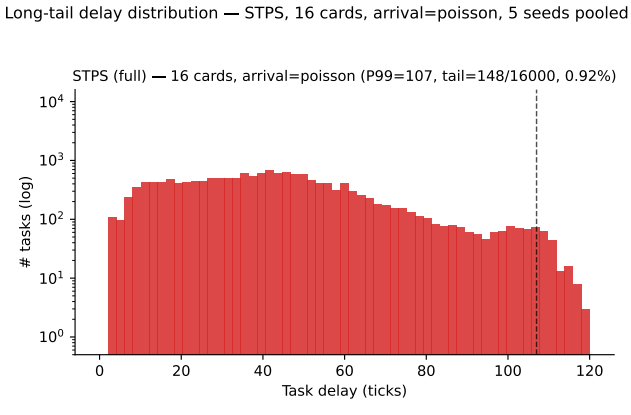
When the trade is favourable. An operator with deadlines tighter than the 6–12 tick p99 gap should run STPS with a smaller $D_{\max}$; the offset-budget sweep (Section 7.4) shows that even $D_{\max}=2$ already keeps the static-baseline tail price under 3 ticks. For workloads whose dominant operational concern is cluster-wide congestion and worst-card load — the typical multi-tenant CIM serving setting we target — the trade is favourable: STPS sacrifices the worst 1 % of tasks to keep the worst card under control, and the worst card is what saturates first.



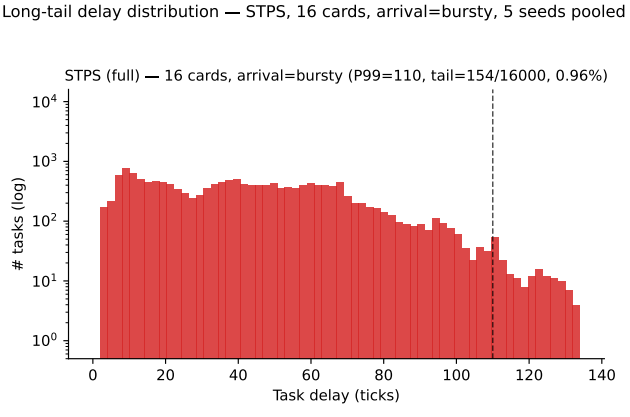
(a) 4 cards, Poisson



(b) 4 cards, Bursty



(c) 16 cards, Poisson



(d) 16 cards, Bursty

Figure 2: Delay survival functions across both cluster sizes and both arrival modes (5 seeds pooled, log–log). All schedulers track each other up to roughly the 90th percentile; STPS's curve only departs from the baselines above $\Pr[\text{delay} > d] \approx 10^{-2}$. The tasks living in that upper-right corner — the ones STPS finishes later — are the same bursty, high-β_τ population that drives p99: STPS pays its tail price exclusively on the long-tail tasks, and pays it to buy the cluster-wide load-balancing and congestion gains documented in Table 1.

8 Conclusion

STPS treats SNN scheduling as a problem of admission-time decisions over a small, physical workload fingerprint, rather than as a per-spike optimisation or a capacity-only bin-packing. Three DTDG-derived quantities suffice to drive two closed-form stages — macro-dispatch and phase-shift — whose combined effect is to keep NoC pressure flat where capacity-based schedulers let bursts collide. In simulation, this trade buys consistent congestion relief at a modest throughput cost, with each stage contributing along a complementary axis. The fingerprint abstraction has room to extend: a fourth quantity, the per-population hotspot indicator, is already extracted but not yet acted on, and the phase-shift kernel assumes a stationary forecast horizon. A scheduler that splits hotspot populations across cards under tail-imbalance pressure, or that re-evaluates the offset choice when the forecast drifts, is a natural next step.

A Cross-Regime Cap-Binding Study

Section 7.4 summarises how phase-shift's effect depends on whether the bandwidth cap binds; this appendix gives the full three-regime sweep behind that summary.

Card-CV reverses sign across regimes, congestion does not. Tables 8 and 9 report the three-regime sweep. The card-CV column tells a clean story: every cell of regime A is negative (phase-shift *raises* card-CV); every cell of regimes B and C is positive (phase-shift *lowers* it). The sign of phase-shift's effect on card-CV is determined entirely by cap binding — not by arrival mode, not by the baseline policy, not even by whether the baseline is a capacity scheduler or STPS-SPATIAL. The congestion ratio, by contrast, moves monotonically: $\downarrow$ in A and C, undefined (already 0) in B. Phase-shift is therefore a congestion lever in the regime where there is congestion to lever and a card-CV reducer in the regime where there is not;

Table 7: Cross-regime cap-binding profile (5 seeds, mean values). The three workpoints differ along a single axis — how heavily the bandwidth cap binds the baseline schedulers. The *offset-infeasible rate* counts tasks for which phase-shift finds no offset in $[0, D_{\max}]$ that keeps the forecast peak below the cap; such a task is still admitted at offset 0 (no task is ever dropped, cf. Section 7.1), but phase-shift cannot relieve its contribution to congestion.

	A: binding 9×10^5, 800	B: non-binding 5×10^6, 800	C: light 9×10^5, 400
Baseline cong. ratio	∼0.26	0.00	∼0.09
Baseline card-CV	0.27–0.29	0.66–0.78	0.65–0.74
Baseline p99 (ticks)	120–137	15–16	24–33
Offset-infeasible (+PS)	0.18–0.22	0.00	0.14–0.16

Table 8: Cross-regime Δcard-CV under +PS (5 seeds, $(off-on)/off \times 100$; positive = phase-shift reduces card-CV). Every cell of the binding regime is negative; every cell of the non-binding and light regimes is positive. The direction of phase-shift's effect on card-CV is determined entirely by whether the cap is binding, independent of arrival mode, baseline policy, or even the spatial policy that phase-shift is attached to.

Arrival	Policy	A: binding	B: non-bind.	C: light
Poisson	RR	−33.7	+25.9	+28.5
	BestFit/DRF	−11.8	+12.7	+25.8
	P2C	−43.9	+27.2	+8.4
	STPS-SPATIAL	−49.7	+19.8	+8.8
Bursty	RR	−34.1	+22.0	+17.8
	BestFit/DRF	−11.8	+12.5	+19.1
	P2C	−45.2	+25.1	+2.8
	STPS-SPATIAL	−47.3	+15.4	+6.7

the underlying mechanism — de-synchronising bursts in time — is the same, but the metric that registers the benefit changes with the workpoint.

The mechanism, broken down. In regime A the cap is already flattening peaks by clipping them; baseline cards therefore look uniform on the served-load axis (card-CV ≈ 0.28), and phase-shift's de-synchronisation moves them off that artificial uniformity, raising card-CV while emptying the queue. In regime B no demand is clipped; phase-shift spreads aligned bursts in time, lowering the peak-to-mean ratio on every card and reducing card-CV directly. Regime C is the intermediate case: enough congestion to give phase-shift something to lever (cong. ratio drops from ∼0.09 to ∼0), not enough to invert the card-CV signal (still −2.8 % to +28.5 %). Notably, the regime B and C numbers include STPS-SPATIAL+ps as a row, which lands at +19.8 % and +8.8 % under Poisson; the Pareto pathology of regime A on STPS+phase (Section 7.5) does not transfer to the regimes where the cap does not bind.

Table 9: Cross-regime direction summary for the remaining Stage-B metrics. Phase-shift relieves congestion only where the cap binds (A, C), and only relieves p99 where the baseline queue already dominates p99 (A). The throughput cost is determined by the offset budget's ratio to the steady-state window, not by the cap.

Metric direction	A	B	C
card-CV	↑	↓	↓
JFI	↓	↑	↑
LIF	↑	↓	↓
Max/min (non-STPS)	↓	↓	↓
Cong. ratio	↓(.26→0)	—	↓(.09→0)
Throughput cost	3–11%	3–10%	14–16%
p99 (non-STPS)	−75/−78	+444/+625	−9/−48
p99 (STPS+ps)	+68/+82	+445/+534	+20

p99 rows: low/high % across seeds and arrival modes.

Extreme congestion is hostile to every scheduler, not just STPS. Read horizontally rather than per-policy, Table 8 delivers a sharper message: in regime A *every cell is negative*, across both arrival modes and across all four spatial policies — RR (−33.7/−34.1 %), BestFit/DRF (−11.8 % twice), P2C (−43.9/−45.2 %), and STPS-SPATIAL (−49.7/−47.3 %). The load-balance regression is therefore not an STPS-specific failing; it is a property of the regime. The same horizontal reading of Table 9 shows the rest of the load-balance panel moving in lockstep — JFI ↓ and LIF ↑ for everyone in column A — while regimes B and C invert every sign. The implication for the operator is direct: STPS's operating sweet-spot is the non-extreme regimes (B, C), where phase-shift's effect on card-CV flips positive across *every* policy and Stage A's placement gains compound with Stage B's de-synchronisation. Once the cluster is pushed deep into regime A, no scheduler in this study — baseline or STPS — improves load-balance under phase-shift; the cap itself has become the flattener, and any further temporal de-synchronisation only re-introduces variance the cap had clipped away. The correct response to regime A is therefore not a smarter scheduler but a provisioning or admission-control decision that moves the cluster out of it; the next paragraph quantifies which of the two interventions is cleaner.

An asymmetry between two ways to relieve the cap. Regime C halves the demand by halving the task count; regime B raises the cap by 5.5×. Both are operationally meaningful ways for a cluster operator to "remove congestion". The offset-infeasible column of Table 7 shows they are not equivalent: halving demand only drops the rate from ∼0.20 to ∼0.15, while raising the cap drives it to zero. Phase-shift's downstream effect on card-CV is consistent with this asymmetry — regime B's improvement (12.5–27.2 %) is larger and more uniform than C's (2.8–28.5 % with wider spread). For an

operator deciding whether to provision more bandwidth or to throttle admissions, the regime sweep argues that bandwidth provisioning is the cleaner intervention for taking advantage of phase-shift.

Acknowledgments

Anonymised for review.

Availability

The simulator, the offline DTDG fingerprint extractor, and the experimental harness used to produce every table and figure in this paper will be released under a permissive open-source licence at publication.

References

[1] Mike Davies, Narayan Srinivasa, Tsung-Han Lin, Gautham Chinya, Yongqiang Cao, Sri Harsha Choday, Georgios Dimou, Prasad Joshi, Nabil Imam, Shweta Jain, et al. Loihi: A neuromorphic manycore processor with on-chip learning. *IEEE Micro*, 38(1):82–99, 2018.

[2] Wei Fang, Yanqi Chen, Jianhao Ding, Zhaofei Yu, Timothée Masquelier, Ding Chen, Liwei Huang, Huihui Zhou, Guoqi Li, and Yonghong Tian. Spikingjelly: An open-source machine learning infrastructure platform for spike-based intelligence. *Science Advances*, 9(40):eadi1480, 2023.

[3] Ali Ghodsi, Matei Zaharia, Benjamin Hindman, Andy Konwinski, Scott Shenker, and Ion Stoica. Dominant resource fairness: Fair allocation of multiple resource types. In *8th USENIX symposium on networked systems design and implementation (NSDI 11)*, 2011.

[4] Robert Grandl, Ganesh Ananthanarayanan, Srikanth Kandula, Sriram Rao, and Aditya Akella. Multi-resource packing for cluster schedulers. *ACM SIGCOMM Computer Communication Review*, 44(4):455–466, 2014.

[5] Arpan Gujarati, Reza Karimi, Safya Alzayat, Wei Hao, Antoine Kaufmann, Ymir Vigfusson, and Jonathan Mace. Serving DNNs like clockwork: Performance predictability from the bottom up. In *14th USENIX Symposium on Operating Systems Design and Implementation (OSDI 20)*, pages 443–462, 2020.

[6] Rajendra K Jain, Dah-Ming W Chiu, William R Hawe, et al. A quantitative measure of fairness and discrimination. *Eastern Research Laboratory, Digital Equipment Corporation, Hudson, MA*, 21(1):2022–2023, 1984.

[7] Sangmin Lee, Rina Panigrahy, Vijayan Prabhakaran, Venugopalan Ramasubramanian, Kunal Talwar, Lincoln Uyeda, and Udi Wieder. Validating heuristics for virtual machines consolidation. *Microsoft Research, MSR-TR-2011-9*, pages 1–14, 2011.

[8] De Ma, Xiaofei Jin, Shichun Sun, Yitao Li, Xundong Wu, Youneng Hu, Fangchao Yang, Huajin Tang, Xiaolei Zhu, Peng Lin, and Gang Pan. Darwin3: A large-scale neuromorphic chip with a novel isa and on-chip learning. *National Science Review*, 2024.

[9] Christian Mayr, Sebastian Höppner, and Steve Furber. Spinnaker 2: A 10 million core processor system for brain simulation and machine learning. *arXiv preprint arXiv:1911.02385*, 2019.

[10] Paul A Merolla, John V Arthur, Rodrigo Alvarez-Icaza, Andrew S Cassidy, Jun Sawada, Filipp Akopyan, Bryan L Jackson, Nabil Imam, Chen Guo, Yutaka Nakamura, et al. A million spiking-neuron integrated circuit with a scalable communication network and interface. *Science*, 345(6197):668–673, 2014.

[11] Jayashree Mohan, Amar Phanishayee, Janardhan Kulkarni, and Vijay Chidambaram. Looking beyond {GPUs} for {DNN} scheduling on {Multi-Tenant} clusters. In *16th USENIX Symposium on Operating Systems Design and Implementation (OSDI 22)*, pages 579–596, 2022.

[12] Deepak Narayanan, Keshav Santhanam, Fiodar Kazhamiaka, Amar Phanishayee, and Matei Zaharia. {Heterogeneity-Aware} cluster scheduling policies for deep learning workloads. In *14th USENIX Symposium on Operating Systems Design and Implementation (OSDI 20)*, pages 481–498, 2020.

[13] Garrick Orchard, E. Paxon Frady, Daniel Ben Dayan Rubin, Sophia Sanborn, Sumit Bam Shrestha, Friedrich T. Sommer, and Mike Davies. Efficient neuromorphic signal processing with loihi 2. In *IEEE Workshop on Signal Processing Systems (SiPS)*, 2021.

[14] Rina Panigrahy, Kunal Talwar, Lincoln Uyeda, and Udi Wieder. Heuristics for vector bin packing. *research. microsoft. com*, 2011.

[15] Sultan Mahmud Sajal, Luke Marshall, Beibin Li, Shandan Zhou, Abhisek Pan, Konstantina Mellou, Deepak Narayanan, Timothy Zhu, David Dion, Thomas Moscibroda, and Ishai Menache. Kerveros: Efficient and scalable cloud admission control. In *17th USENIX Symposium on Operating Systems Design and Implementation (OSDI 23)*, pages 227–245, Boston, MA, July 2023. USENIX Association.

[16] Qizhen Weng, Lingyun Yang, Yinghao Yu, Wei Wang, Xiaochuan Tang, Guodong Yang, and Liping Zhang. Beware of fragmentation: Scheduling {GPU-Sharing} workloads with fragmentation gradient descent. In *2023 USENIX Annual Technical Conference (USENIX ATC 23)*, pages 995–1008, 2023.

[17] Wencong Xiao, Romil Bhardwaj, Ramachandran Ramjee, Muthian Sivathanu, Nipun Kwatra, Zhenhua Han, Pratyush Patel, Xuan Peng, Hanyu Zhao, Quanlu Zhang, et al. Gandiva: Introspective cluster scheduling for deep learning. In *13th USENIX Symposium on Operating Systems Design and Implementation (OSDI 18)*, pages 595–610, 2018.

[18] Zhaokun Zhou, Yuesheng Zhu, Chao He, Yaowei Wang, Shuicheng Yan, Yonghong Tian, and Li Yuan. Spikformer: When spiking neural network meets transformer. In *International Conference on Learning Representations (ICLR)*, 2023.